

THE ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
MODULE 1 LEVEL(3,4,5&6)

ELEMENT COVERED: PERFORM TROUBLESHOOTING
UNIT OF COMPETENCE: COMPUTER REPAIR AND MAINTENANCE
Duration: 2 Hours
Total Marks: 50

NAME.....DATE.....

CDACC REGNO.....SIGN.....

- **INSTRUCTIONS**
- This is a **hands-on practical assessment**. You must physically perform all tasks.
- Do not guess answers—marks are based on actual actions and observations.
- Handle all computer components carefully to avoid damage.
- Follow proper **troubleshooting procedures step by step**.
- Write all findings clearly in your answer sheet.
- Do not open components unnecessarily unless required.
- You may power the system on/off during testing.
- Work independently .

1. MATERIALS / RESOURCES REQUIRED

Each candidate will be provided with the following:

A. Hardware Equipment

- 1 Desktop computer system (intentionally faulty or pre-configured fault)
- Monitor (VGA/HDMI display unit)
- Keyboard
- Mouse
- CPU casing (opened or accessible for troubleshooting)
- Power supply unit (PSU)
- RAM modules (at least 1–2 sticks installed)
- Hard disk or SSD

- Power cables
- Data cables (SATA/VGA/HDMI)

B. Tools for Troubleshooting

- Screwdriver set (flat and Phillips)
- Anti-static wrist strap
- Multimeter
- Flashlight

C. Documentation Materials

- Answer booklet / writing paper
- Fault report template (provided by examiner or written by candidate)
- Pen and pencil

D. Software/BIOS Access (if available)

- BIOS/UEFI interface
- Boot diagnostic screen
- Pre-installed OS

SCENARIO

A technician in Aberdare technical has a desktop computer in ICT lab that is faulty. It has the following issues:

- No display
- No power
- System fails to boot
- Peripheral failure

You are required to:

- Assess the user problem
- Identify the fault
- Diagnose the cause
- Repair the system
- Document your solution

TASK 1: USER DATA ASSESSMENT

Instructions:

1. Interview the user (examiner acting as user).
2. Ask about:
 - Nature of problem
 - When it started
 - Recent changes made
 - Symptoms experienced
3. Write down:
 - User details
 - Device location
 - Full problem description
4. Summarize the fault in one sentence.

TASK 2: COMPUTER PROBLEM IDENTIFICATION

Instructions:

1. Inspect the system physically:
 - Check power connection
 - Check monitor connection
 - Observe system lights and fans
2. Power on the computer and observe:
 - Boot process
 - Beep sounds
 - Error messages
 - Display status
3. Open the CPU casing and check:
 - RAM seating
 - Hard disk/SSD cables
 - Power supply connections
 - Loose internal cables

4. Record all findings clearly.

TASK 3: ANALYSIS OF USER DATA

Instructions:

1. Compare:
 - User complaint
 - Physical inspection results
2. Identify possible cause(s):
3. Write a clear hypothesis explaining:
 - What you think is wrong
 - Why you think so

TASK 4: SOLUTION AND REPAIR

Instructions:

1. Apply corrective actions such as:
2. Power on the system again.
3. Test:
 - Boot process
 - Display output
 - Keyboard/mouse response
4. Record whether the system is working or not.

TASK 5: DOCUMENTATION OF SOLUTION

Instructions:

Write a final report including:

- Fault identified
- Cause of the fault
- Steps taken to solve it
- Final system status
- Recommendation for prevention

The trainer is introducing oral questions shortly for one hour